CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in machine learning and deep learning, from model training and evaluation to deployment, with image classification (CNN, ResNet) project work and RAG and agentic AI systems in Python. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by a strong foundation in data analysis and software development across enterprise and start-up environments.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer owning an agentic GenAI knowledge-extraction system end-to-end: data ingestion, chunking, prompt engineering and indexing (agentic, deterministic and monolithic strategies), reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Designed and optimized multi-agent LLM workflows with LangGraph (entity-extraction, relationship-analysis, consolidation and reflection agents running in parallel) on complex regulatory and technical documents, with evaluation against an SAP-system-derived ground truth raising precision/recall from 33%/45% to 93%/90%.
- Designed and implemented a Model Context Protocol (MCP) query service that connects the knowledge graph to AI agents with millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) deployed on Azure, where domain experts review and approve knowledge-graph changes before they are applied.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Delivered with an agent-native approach as an AI-augmented engineer, using Claude Code, custom skills and agentic coding workflows for rapid, high-quality development grounded in clean code principles.
- **Tech stack: Python, LangGraph, MCP, multi-agent systems, LangChain, LiteLLM, RAG, GraphRAG, FastAPI, REST APIs, microservices, Docker, Kubernetes, Azure, Postgres, Neo4j, MongoDB, React**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Worked with business stakeholders to map their decision-making workflows and translated the requirements into an agentic AI application with LangGraph, deployed to AWS, that routes each request across vector database queries, general LLM inference and summarization.
- Developed a Python backend with FastAPI REST APIs and a React front-end, using Postman for API testing.
- Built and containerized the entire stack with Docker and deployed it to AWS using CDK (infrastructure as code), following clean code practices; versioning in GitLab with CI/CD deployment.
- **Tech stack: Python, FastAPI, LangGraph, LangChain, React, Postman, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated LLM evaluation of RAG generation and retrieval with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing, and fine-tuning models for optimized retrieval.
- Built AWS cloud infrastructure with Lambda, SQS and DynamoDB for PrismaChat RAG evaluation using CDK.

- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- **Tech stack: Python, LangChain, Hugging Face Transformers, PyTorch, RAGAS, PytrecEval, AWS (Lambda, SQS, DynamoDB, CDK), Docker**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Engineered lag features and rolling statistical windows from raw sensor time-series (temperature, reaction time, process variables) to feed the tree-based models, while feeding raw sequences directly into the LSTM.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Machine Learning, Deep Learning, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, SciPy, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XG-Boost regressor, FB Prophet, LSTM models for time series analysis for EVs.
- Developed LSTM, achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Image Defect Classification Pipeline | Python | PyTorch | FastAPI | Docker

- Fine-tunes a pretrained image model in PyTorch on public road-damage street images (RDD2022: cracks, potholes), with training data quality checks, a training and evaluation pipeline reporting per-class precision and recall, and a FastAPI inference service in Docker with CI.

Project: Point Cloud Processing Pipeline | Python | Open3D | NumPy | FastAPI

- 3D image processing pipeline for public point cloud scans (e.g. ModelNet, S3DIS): loading, voxel downsampling, outlier removal, normal estimation and RANSAC plane segmentation algorithms, with a FastAPI endpoint and a small baseline classifier evaluated per class.

TECHNICAL SKILLSET

- **Machine Learning & Deep Learning:** PyTorch, TensorFlow/Keras, Scikit-Learn, XGBoost, model training, validation and evaluation, time-series models (LSTM), QLoRA fine-tuning, Hugging Face Transformers
- **Computer Vision & 3D (project work):** CNN image classification, Open3D point cloud processing, NumPy, SciPy, Matplotlib
- **Software Development:** Python, FastAPI, REST APIs, Postman, Git, GitLab CI/CD, Docker, clean code, Agile
- **GenAI & Data:** LangGraph, LangChain, RAG, MCP, Pandas, SQL, Postgres, Vector Databases, Streamlit
- **Languages:** English (Business Fluent), German (B1), Hindi (Fluent), Telugu (Native)

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering (Minor in Computer Science & Engineering) — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles